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Introduction:
Generative adversarial networks which are known as GAN show a significant breakthrough in generative artificial intelligence. The two neural networks pitting concepts against each other in a zero sum game which are generator and discriminator which distinguish between real and fake data GANs can learn to analyse patterns between complex data distribution. This report explores various types of the versatility and architectural requirements of GANs under the three domains which are 2D continuous data mathematical curves , Microscopic Medical Imaging and cybersecurity network data. The case study explores how the adversarial network adapted to specific data structures, the unique training challenges inherent to each domain. These are the methods evaluating synthetic data quality.

Objective and Target Distribution
The fundamental challenge of the distribution of the standard GAN is mapping a random noisy vector to a high non-linear target distribution.
The experiment where the noisy curve explains the curve defined by the equation $y=\sin(2x)+0.3\cos(5x)+e$ where x belongs to [-5,5]and $e \sim N(0,0.1)$.the distribution presents specific challenges: the high frequencies is cosine wave creates rapid fluctuations and the gaussian noise where the curve is statistical thick rather than straight distinct line.

Architectural Adaptions
Standard Multilayer preceprtron (MLPs)were utilised by generators and discriminator. To capture the complex geometry of the targeted curve, the standard base line architecture was expanded to three hidden layer with the wider capacity upto 128 neurons

Activation : the activation are critical in preventing the dying relu problem,ensuing the gradients continued to flow even when the neural network produced negative outputs therefore stabilising the adversrail training process

Dimensionality: the generator tools a 2D noise vector as input and output as 2D co-ordinate(x,y)

Evaluation and Failure Modes
The evaluation in 2 dimensions is extremely intuitive and strongly depends on the visual scatter plot. In particular, a correctly trained model draws the generated points in perfect alignment along the way of the true math function, with exact variance from the noise term replicated. While training, one of the primary issues that appeared is the mode collapse phenomenon. In this case, the Generator finds one coordinate that deceives the Discriminator into believing that this is the only possible outcome, generating this coordinate again and again without mapping the entire curve at all.
Plot referenced in the fig-1

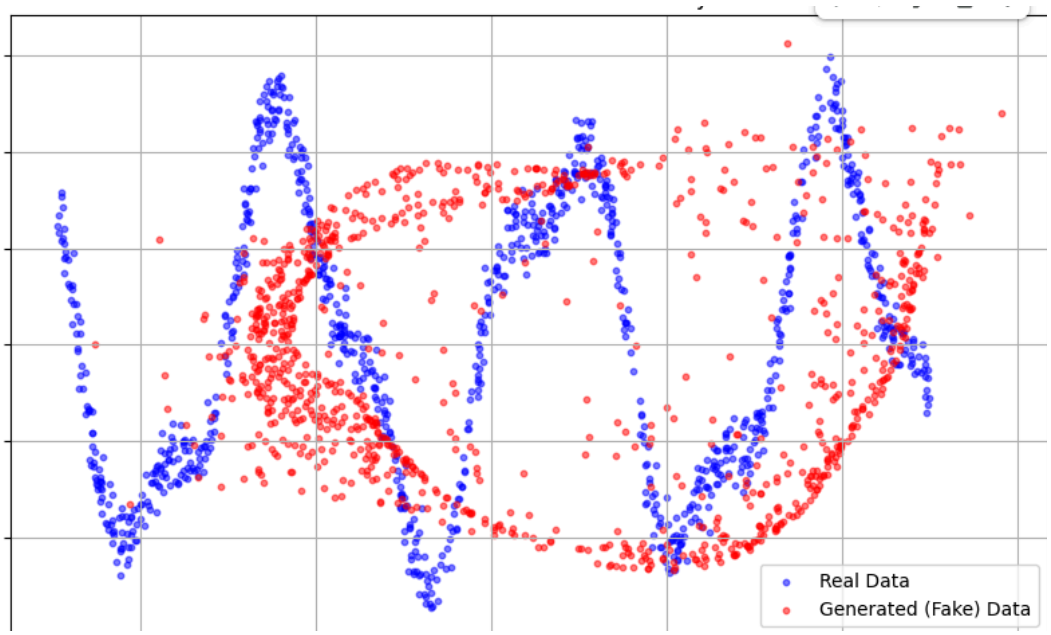

Fig1

Medical Image Synthesis with BloodMNIST

Spatial Data and the DCGAN Architecture

The process of moving from coordinate representation to image generation necessitates an upgrade from basic MLP models to DCGANs. The BloodMNIST database, consisting of 28x28 pixel images of eight different blood cell types, was used for training purposes. To conform to the design of regular DCGANs, these images were scaled up to 32x32 pixels.

Generator: Employed ConvTranspose2d for upsampling from a 100 dimensional noise vector to a 32 x 32 x 3 RGB image. This worked like an inverse classifier.

Discriminator: Used regular Conv2d layers with stride for downscaling the image and getting spatial features out to ascertain whether the image is authentic.

Normalization: BatchNorm2d was employed extensively to keep the distribution mathematically stable, so that no one pixel or channel overwhelms the whole learning process.

Quantitative Evaluation: Fréchet Inception Distance (FID)

Human observation of the blood cell images can be considered useful, but it is rather subjective. Quantitative analysis was performed using Fréchet Inception Distance (FID). FID works by feeding both real and fake images into a pre-trained model (InceptionV3). The process involves measuring the distance between the two multivariate Gaussian distributions obtained from the deep feature maps produced by the pre-trained model. A low FID score suggests that the fake images have similar edge details, textures, and colors as those in the real dataset.

Reflection:
DCGAN excelled at producing realistic blood cells quickly for BloodMNIST. Yet, this high performance is entirely dependent on the characteristics of the dataset: the blood cells are essentially spheres, uniformly centered, and placed on the uniform background of the microscopic slide.
This same network, when applied to generate real-life duck images, would produce very low-quality outputs. Ducks have a complex body structure with several components (head, beak, wings, eyes, etc.), which need to be coherent. Ducks live in a messy and unpredictable environment. DCGAN may only produce blobs with multiple heads without further training on a much bigger database and much deeper neural network architecture.

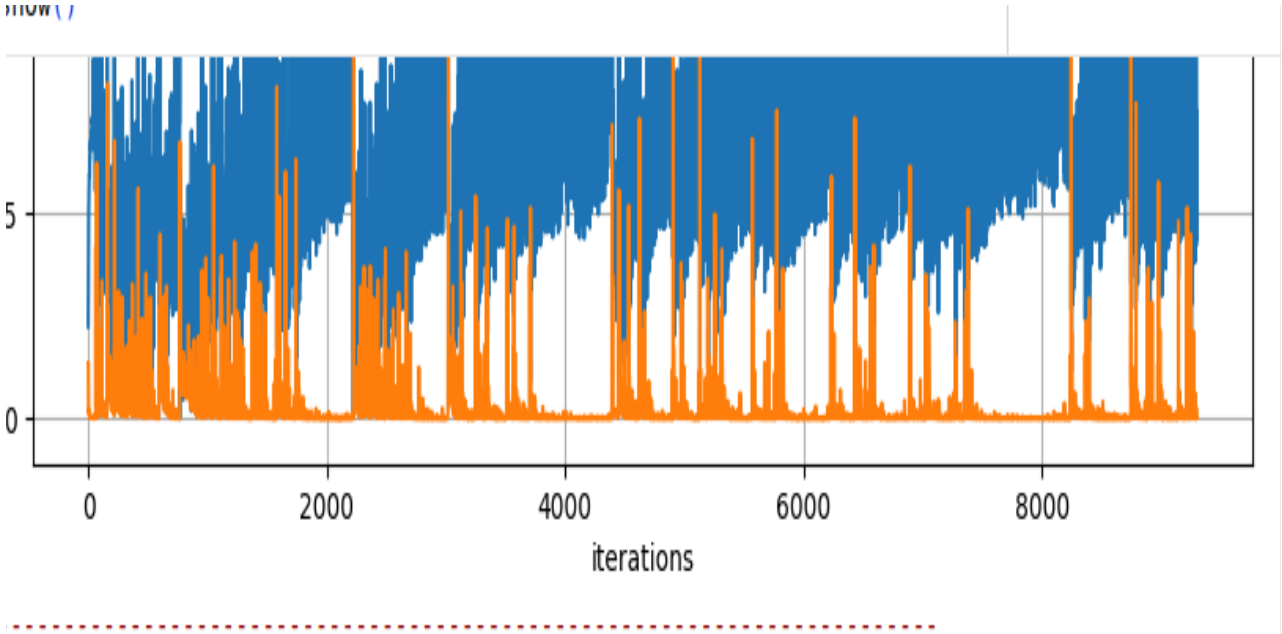

Fig-2

Cybersecurity

Creation of artificial traffic for cybersecurity purposes entails difficulties that do not arise when creating artificial images or artificial mathematical models. The CICIDS 2017 data set contains tabular rows with 78 different network attributes Flow Duration, Destination Port, Packet Length.

Data cleaning: Neural networks do not handle NaN or Inf data very well; however, this was common in our raw captures from the network. They needed to be cleaned by dropping and replacing NaNs/Inf values.

Scaling: Our network features differ widely in scale (Port number: 80, Flow duration: Millions). We had to aggressively scale our data using MinMaxScaler(-1, 1) because our activation function at the Generator output is Tanh.

Architecture and Regularization

Since network features do not have spatial relations (as compared to images), Convolutional layers were ineffective. The architecture adopted the 1D Linear layer (as opposed to the 2D data experiment). However, a lot of capacity was needed to cope with 78 input/output dimensions.

Dropout: Dropout layers were incorporated into the Discriminator. As a means of regularization, dropout randomizes a fraction of the neurons. It ensures that the Discriminator does not simply memorize the exact network traffic fingerprints but learns generic patterns.

PCA-Based Visualization and Validation of Results

It is impossible for humans to evaluate 78-dimensional data. To visually validate the effectiveness of the generated results, Principal Component Analysis (PCA) was used. PCA used the 78-dimensional data and calculated only two principal components (X and Y). They represent the maximum variance contained within the dataset. We used PCA transformation for both the actual and generated traffic, which enabled us to visually analyze how much overlap between the two sets there was. A good quality tabular GAN should generate samples with distribution exactly following the clusters of actual traffic.

Visual and Evaluation Through PCA

Human analysts cannot evaluate the 78-dimensional dataset. In order to see if the GAN is working effectively, PCA was used. Using the mathematical approach of PCA, the 78 dimensions were brought down to two key dimensions known as the X and Y axes, which capture all the variance present in the data. Comparing the PCA-based real traffic data set with the PCA-based synthetic data set can show if there is any overlap between them. If the GAN works effectively, then it will result in perfect tracing of the shapes of benign and DoS traffic.

Generalization and Concept Drift

Although the GAN was able to synthesize realistic DoS attack traffic using the Wednesday dataset, generalization to any other form of attack traffic beyond that which it was trained on is not possible. A GAN that is only trained on DoS attacks would not be able to synthesize realistic PortScan or Web Attack attack traffic at all. Additionally, network topologies and attack techniques experience high rates of concept drift. Synthetic data based on 2017 datasets would be useful for research purposes but would not be suitable for testing new IDSs in real time against zero-day attacks.

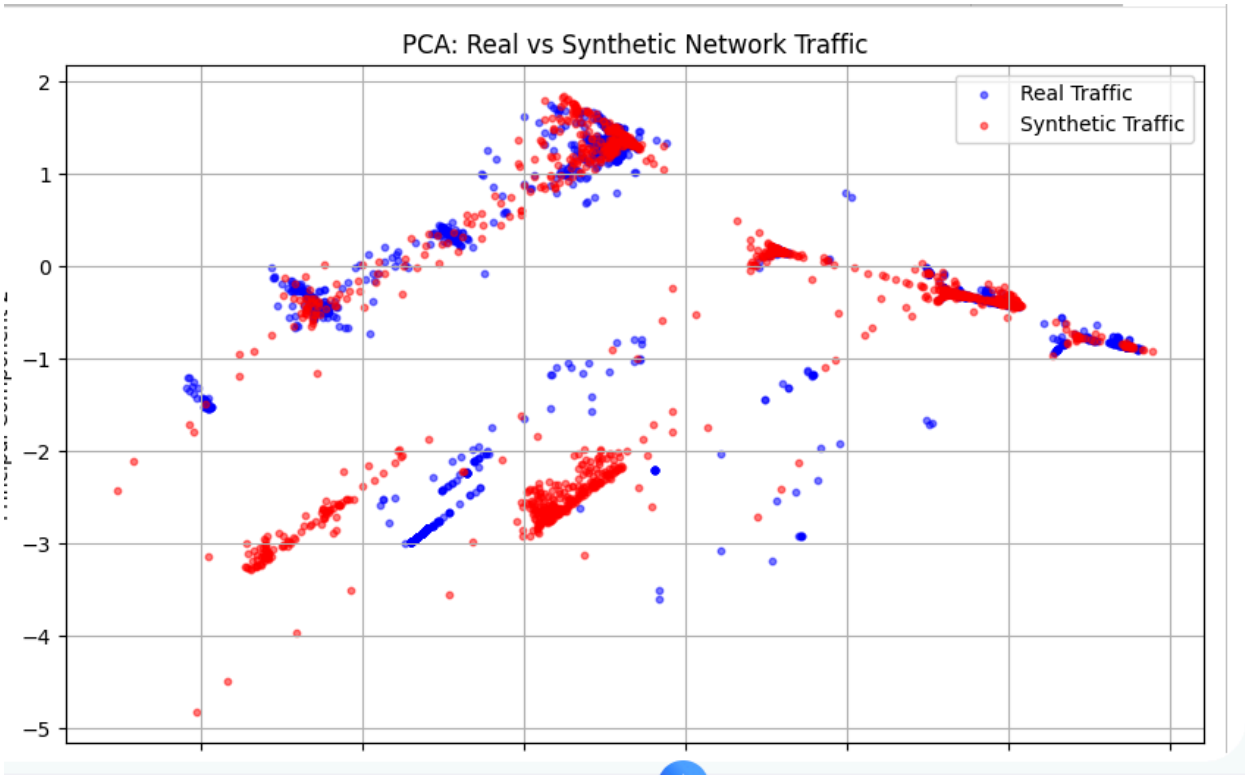

Fig 3

Quick Draw Pizza

Training a DCGAN on the QuickDraw "pizza" dataset will force the model to construct sparse pen strokes rather than the dense, realistic image representations that come from other kinds of datasets. This involves training a generator, which constructs images with transposed convolutions, against a discriminator that can detect fake images in an unstable learning environment where balancing the loss metrics is critical to avoid problems such as mode collapse. The effectiveness of the constructed pizzas is then verified both visually and quantitatively by measuring the Fréchet Inception Distance (FID), which confirms whether the images correspond to drawings done by humans. Despite its effectiveness in constructing simple figures such as pizzas and houses, the DCGAN architecture finds it difficult to create figures with a lot of variation, such as cats.

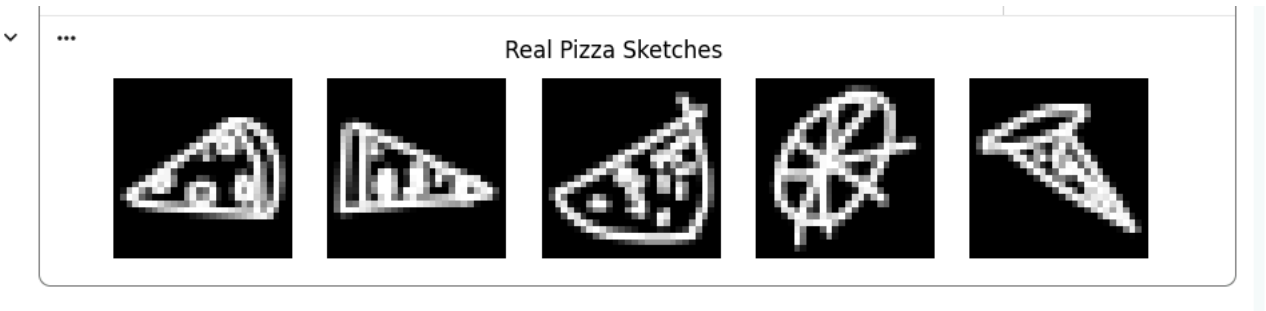

Fig 4

Conclusion

Although Generative Adversarial Networks offer a versatile and adaptable approach to synthetic data generation, there is no “one size fits all” model here. The purpose of this report is to show that although the adversarial loss function does not change throughout, the actual neural networks employed, the pre-processing procedures carried out and the metrics used to evaluate the generated data have to be completely rethought in accordance with the particular field of study. Mathematical functions call for large MLPs due to frequent fluctuations, images – convolution layers because of spatial structure mapping, and cybersecurity tabular data needs rigorous feature selection and dropout due to the discrete nature of the information processed.